PRODUCT MANAGER CASE STUDY · RECRUITMENT MARKETPLACE

Unified Tagging System

One tagging layer that lets employers and job seekers categorise, prioritise and track everything on the platform, on any device, alone or as a team.

#flexible #scalable #smart #adaptable #cross-device #collaborative #data-driven

Prepared by Ibrahim · November 2026

SUMMARY

The answer on one slide

PROBLEM

Too much to track, no shared way to track it

Jobs, candidates and applications keep growing. Pipeline stages are too rigid and private notes don’t roll up, so people can’t organise their way and managers can’t report.

SOLUTION

One tag engine, three scopes

Platform, team and personal tags on any object, served by a headless service. Suggestions reuse before they create, and light governance keeps the list short.

HOW WE’LL KNOW

Measure recall, not just creation

North Star: tag-powered recalls per weekly active user. Backed by adoption, suggestion acceptance, time to tag, clutter ratio and p95 latency.

PLAN

Four phases over two quarters

Personal tags MVP, then team tags and governance, then smart suggestions and analytics, then platform insights. Each phase ships behind exit criteria.

AGENDA

How this deck answers the brief

Task	From the brief	Slides	What you'll see
1	Research & analysis	4–8	Research plan, four personas, challenge map, scope
2	Design	9–17	Principles, tag model, architecture, AI, governance
3	Prototyping	18–21	Clickable prototype on desktop, tablet and mobile
4	Analytics & reporting	22–26	North Star, metric targets, event schema, review cadence
5	Feedback & iteration	27–28	User tests by phase and the feedback loop
6	Performance & scalability	29–30	Design targets, SLOs, load and relevance testing
+	Risks and roadmap	31–33	Compliance risks, phased plan, summary

1 · RESEARCH & ANALYSIS

Four lenses before we design anything

User interviews

16 sessions, 4 per persona segment. How do people remember, sort and hand off items today?

Behaviour data

Repeated searches, saved-item revisits, notes usage and CSV exports. Each is a proxy for a missing way to organise.

Support and sales

Ticket and lost-deal themes: “can’t find”, “can’t share”, “can’t report”, “too many labels”.

Competitive teardown

Applicant tracking systems for team workflows; Gmail labels, Notion and Linear for fast personal tagging.

No platform data came with the brief, so the insights that follow are hypotheses. Each one names the signal that would confirm or kill it.

1 · RESEARCH & ANALYSIS

Four personas: solo or team, hiring or job hunting

Employer · solo

Sam, owns a 30-person bakery chain

Needs: remember who was good last time; hires 2–3 roles a year.
Tags give: private shortlists and silver-medalist recall for the next opening.

Employer · team

Priya, talent acquisition lead

Needs: keep 30 open reqs moving with 5 recruiters and 20 hiring managers, then report upward.
Tags give: governed team tags, shared queues, tag analytics.

Job seeker · solo

Daniel, frontend developer

Needs: know where he stands on 40+ saved jobs, mostly from his phone.
Tags give: automatic status tags, follow-up reminders, one-tap capture.

Job seeker · team

Maya, university career advisor

Needs: guide a cohort of 60 graduates without micromanaging.
Tags give: tags students choose to share, plus a “coach review” queue.

Plus one reader in every company: the VP who never tags anything but needs Priya’s tags to roll up into a report.

1 · RESEARCH & ANALYSIS

Where users struggle, and where tags help

Challenge in the brief	What users likely do today	How UTS responds
Volume and diversity	Re-run searches, scroll, keep spreadsheets	Tag once, recall with one click or a saved view
Growth and clutter	Labels multiply until nobody finds the right one	Reuse-first suggestions, synonyms, merge and archive
Solo and team users	One rigid workflow forced on everyone	Three scopes: personal, team, platform
Collaboration	“Can you look at this one?” over chat and email	Shared tags as hand-offs, views as queues
Changing recruitment	Every new workflow waits for engineering	Tag groups and rules are configuration, not schema
Many devices	Power features only on desktop	One picker component that adapts per device
Data-driven insights	Export to CSV and pivot by hand	Tag events feed customer reports and our roadmap

THE CORE TENSION

Individuals need freedom to label things their own way. Managers need structure to report on them.

So UTS is **personal by default**, **shared by intent** and **governed at the top**.

1 · RESEARCH & ANALYSIS

What UTS is, and what it isn’t

UTS is

- A light classification layer on any object: jobs, candidates, applications, profiles, companies, saved searches
- Private or shared, by the user’s choice
- Filterable, saveable as views, and reportable
- One service every surface and partner can call

UTS is not

- A replacement for pipeline stages; workflow state stays in the stage model
- A notes or comments tool
- A place for judgements about protected characteristics; those terms are blocked
- A taxonomy project that must be finished before launch

MVP order: candidates and jobs first, because they carry the most volume. Applications, companies and saved searches follow on the same engine.

2 · DESIGN

Five principles every decision is tested against

Two taps to tag	Capturing must cost less than remembering, or nobody does it.
Reuse before create	Suggestions and synonyms come first. Creating a new tag is the last option.
Personal by default	One person’s shorthand never clutters the team list. Sharing is a deliberate step.
Govern groups, not tags	Admins lock the few groups reports depend on, like Status, and leave the rest open.
One engine, many surfaces	The tag service knows nothing about screens, so screens can change freely.

2 · DESIGN

The tag model: three scopes, typed groups

Platform #open-to-remote #visa-sponsorship

Defined by us for everyone. Powers marketplace-wide filters and benchmarks. Some are applied automatically, like #applied.

Team #hm-review #strong-yes

Created by members, governed by the org admin. Visible to the team; feeds shared views and reports.

Personal #top-pick #call-back

Anyone, any time, visible only to its owner. Can be promoted to Team when it proves useful.

EVERY TAG CARRIES

```
name · display label
scope · platform | team | personal
group · status, priority, skill
aliases · “js” → javascript
entity types · where it applies
owner · org · created by
state · active | archived | merged
usage · count, last used
```

On screen, colour shows the group and a lock icon marks personal tags, so visibility never depends on colour alone.

2 · DESIGN

Architecture: a headless tag service

SURFACES

- **Web app**
- **iOS, Android**
- **Tablet**
- **Email, extension**
- **Customer API**

↓ One shared TagPicker component and design tokens · REST and GraphQL gateway

TAG SERVICE

- **Tags**
  create, alias, merge, lock
- **Taggings**
  any object, bulk jobs
- **Suggest**
  rank, search, typo-tolerant
- **Rules**
  notify, auto-tag, webhooks

↓ Writes go to the store of record; every change is also published as an event

DATA AND LEARNING

- **Postgres**
  source of truth, by tenant
- **Search index**
  prefix and fuzzy match
- **Redis**
  hot tags per user
- **Event stream**
  warehouse and models

2 · DESIGN

Staying adaptable as recruitment changes

Entity-agnostic taggings

A tagging is just tag, object type, object id, actor and time. New objects need no migration.

Configuration, not code

Groups, locked tags and rules like “#hm-review notifies the hiring manager” are set per org.

Server-driven picker

The API returns sections and reasons; clients only render them. We change ranking without app releases.

Flags and experiments everywhere

Every new flow ships dark, then to a cohort, then to all.

THE CONTRACT EVERY SURFACE USES

```
GET /v1/suggest
  ?entity=candidate:812&q=js

{ "sections": [{
    "title": "Matches",
    "items": [{
      "tag": "javascript",
      "scope": "team",
      "reason": "synonym:js" }]
  }],
  "allow_create": false }
```

2 · DESIGN

Smart suggestions: rank five signals, show five tags

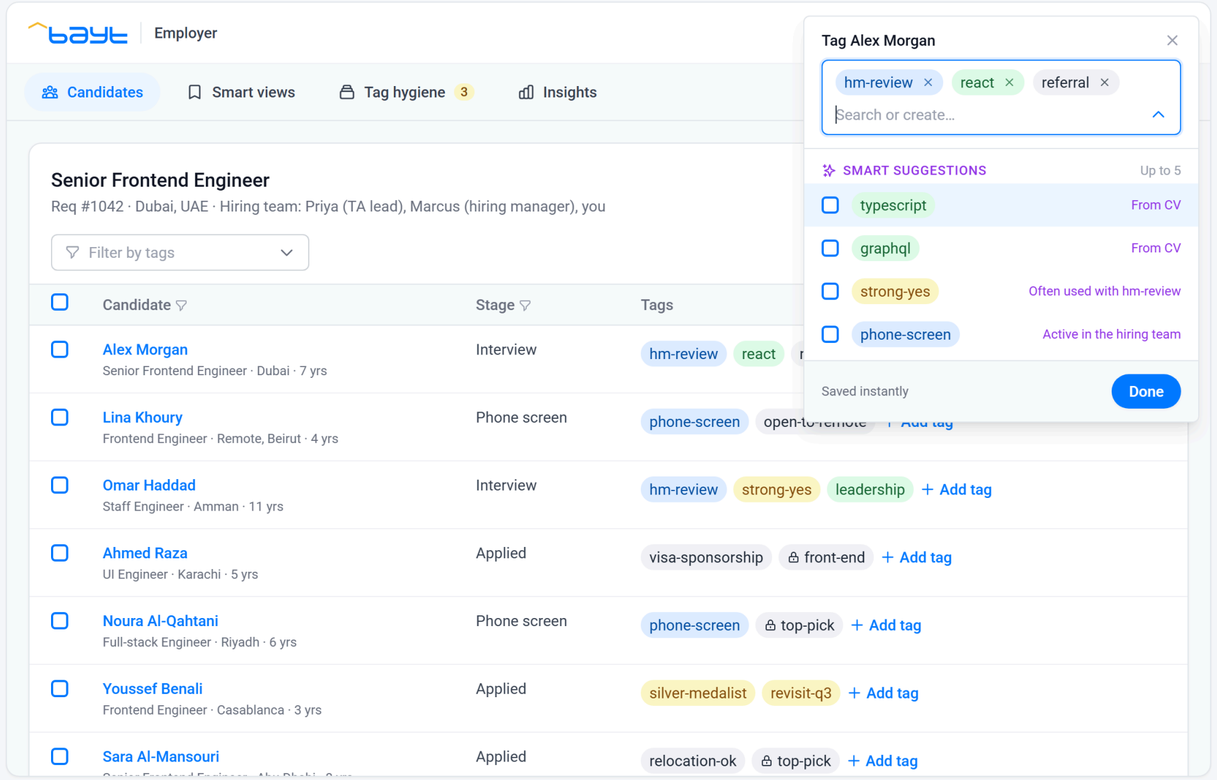
Content
Skills and keywords from the résumé or job post

Co-usage
Tags this team usually applies together

Your habits
Recency and frequency of your own tags

Team activity
Tags active in this req or workspace this week

Context
Object type, pipeline stage and device



Every suggestion shows its reason. Version 1 uses weighted rules; phase 3 learns the weights from which suggestions people accept.

2 · DESIGN

AI in tagging: AI suggests, a person decides

AI feature	What it does	Why it helps
Auto-tag from CVs	Reads CVs and job posts, proposes skill tags as dashed chips	Hours of manual tagging become one click
Smart suggestions	Ranks the five most likely tags, each with a reason	Faster tagging, fewer new tags
Ask in plain words	“remote react” becomes the tags open-to-remote and react	No need to learn the tag list
AI clean-up	Finds duplicates, unused tags and popular personal tags	Clutter never builds up
AI summary	Turns tag data into two or three plain sentences	Managers see bottlenecks at a glance

GUARDRAILS

AI never tags on its own: a person accepts each tag or the batch, and can undo it. AI tags are marked as AI. Protected characteristics are blocked. We track how often AI tags are accepted and how often they are removed.

2 · DESIGN

Avoiding clutter when an org has 10,000 tags

Prevent

At most five suggestions. Synonyms and typo matching run before “Create”; near-duplicates get a warning.

Detect

A nightly job flags near-duplicates and tags unused for 90 days.

Resolve

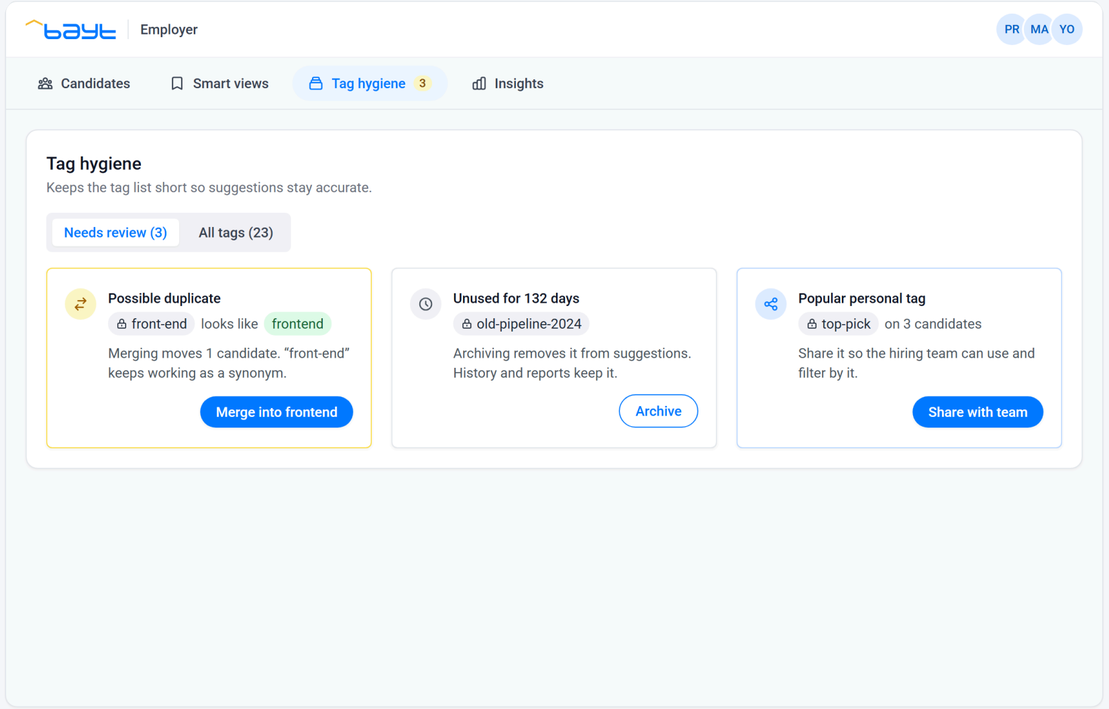
Merge keeps the old name as a synonym, so no view breaks. Archive keeps history.

Promote

Often-used personal tags are offered to the team instead of being reinvented.

Govern

Admins lock report-critical groups and decide who creates team tags. Every change is audited.



2 · DESIGN

Collaboration: a shared tag is a hand-off

Recruiter

Priya tags a candidate

#hm-review

→

Hiring manager

Marcus's review queue updates; daily digest

#strong-yes

→

Coordinator

"Ready to schedule" view picks it up

#onsite

→

Leadership

Report: time in each tag, hand-off delay

Permissions

Viewer sees and filters by team tags.
Member applies team tags and keeps personal ones.
Admin creates, merges and locks team tags, and sets rules.

Job seekers work in teams too

Daniel tags a job #coach-review and it appears in Maya's queue. Only tags he chooses to share are visible to her; his personal tags stay private.

2 · DESIGN

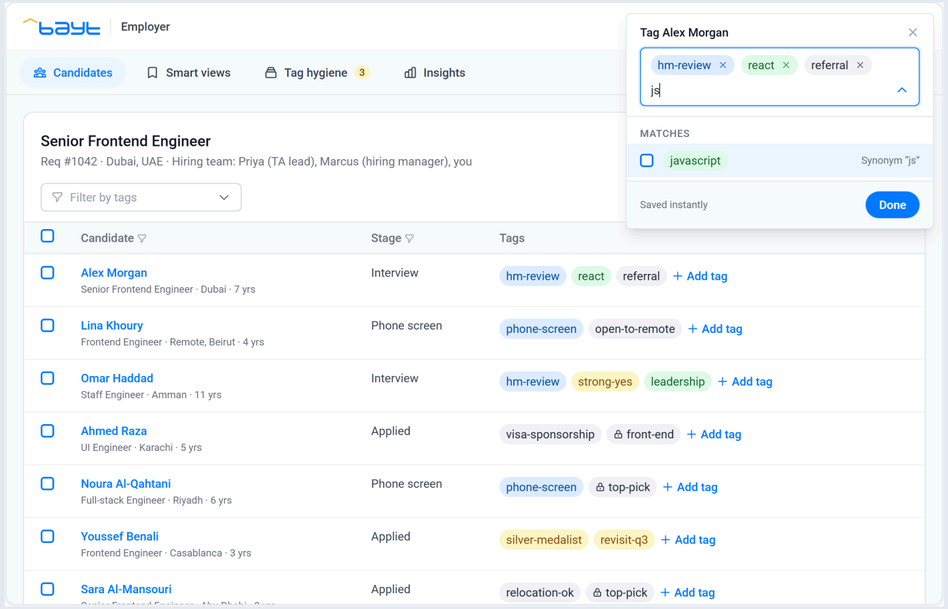
Six core flows, each with a target

Flow	Steps	Target
Quick tag	Open picker, pick a suggestion	2 taps · median under 3 s
Bulk tag	Select rows, press T or tap “Tag”, pick	Same 3 steps for 3 or 3,000 items
Recall	Click a tag chip, or open a saved view	1 click · results p95 under 500 ms
Hand off	Apply a team tag that feeds a queue	Seen by the owner within a day
Share a tag	Promote a personal tag to the team	2 taps · no re-tagging needed
Clean up	Admin merges or archives from hygiene	1 click · undoable · no broken views

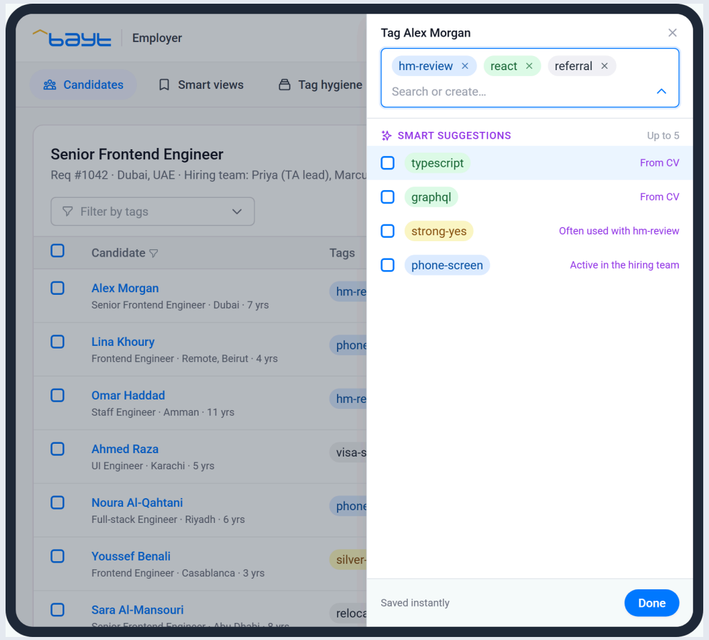
Every flow ends with an undo toast, and every flow works the same on desktop, tablet and mobile.

3 · PROTOTYPING

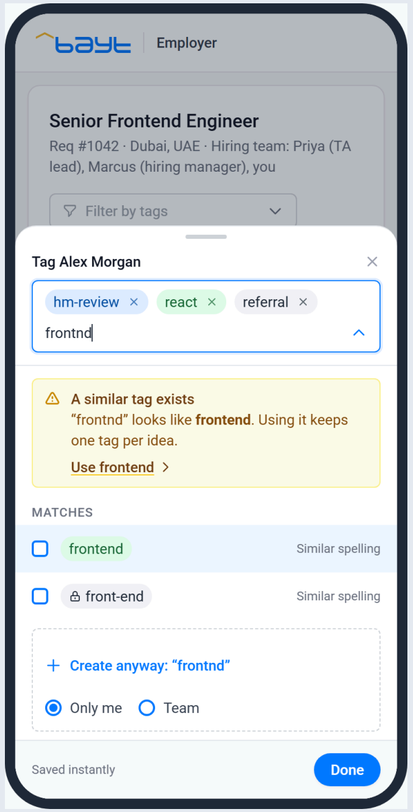
One picker component, three devices



Desktop: popover, keyboard first



Tablet: side sheet

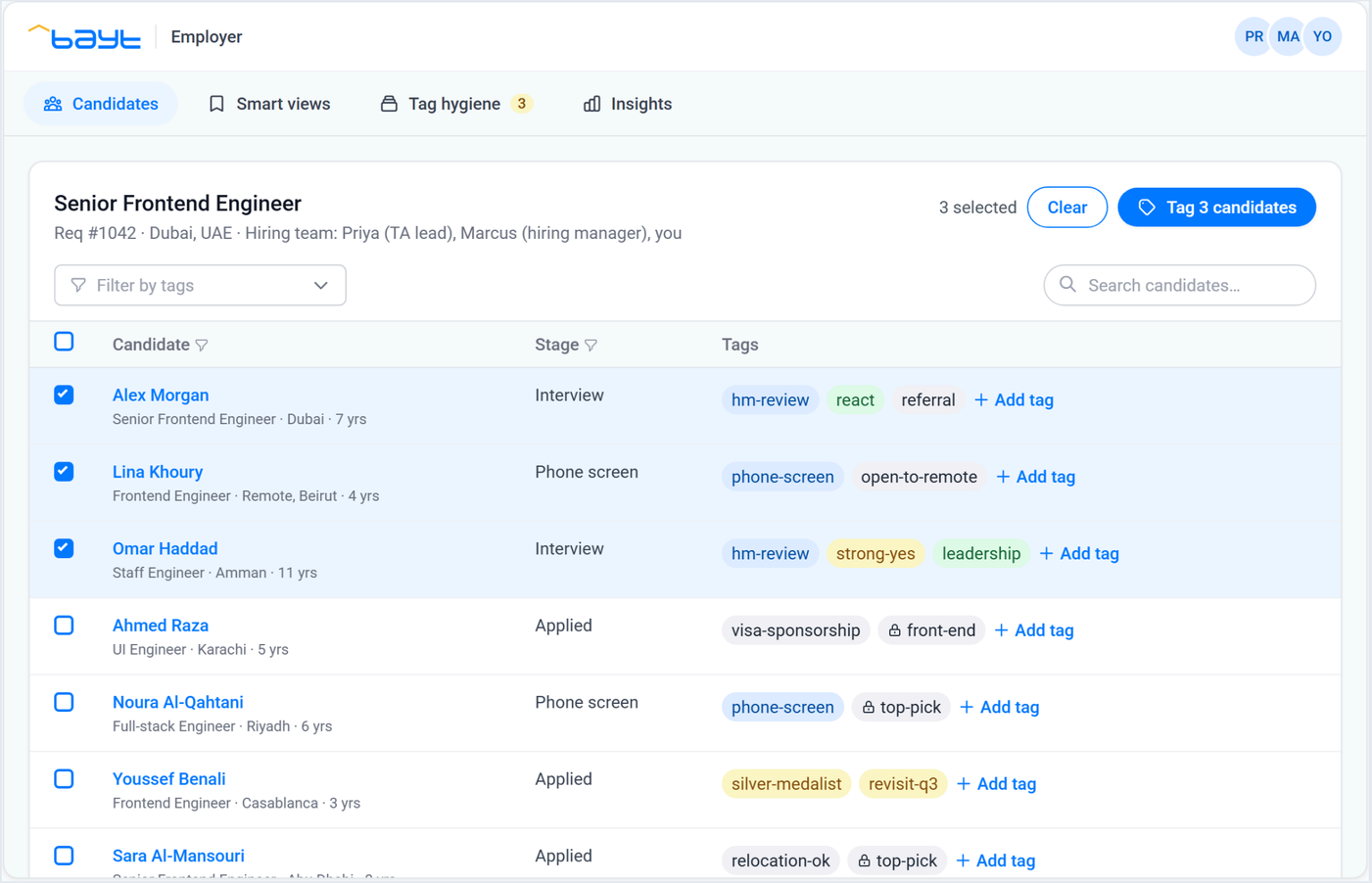


Mobile: bottom sheet

Try it: mibbrahim.github.io/unified-tagging-system · switch device and persona at the top

3 · PROTOTYPING

Desktop: built for volume



Every tag is a filter

Click a chip in any row to narrow the list, then save it as a view.

Bulk without a new screen

Tick rows and the same picker tags all of them.

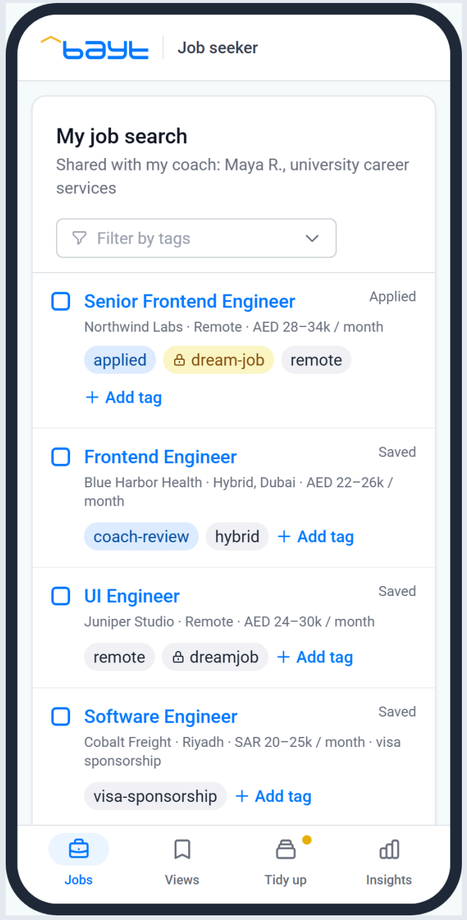
Keyboard first

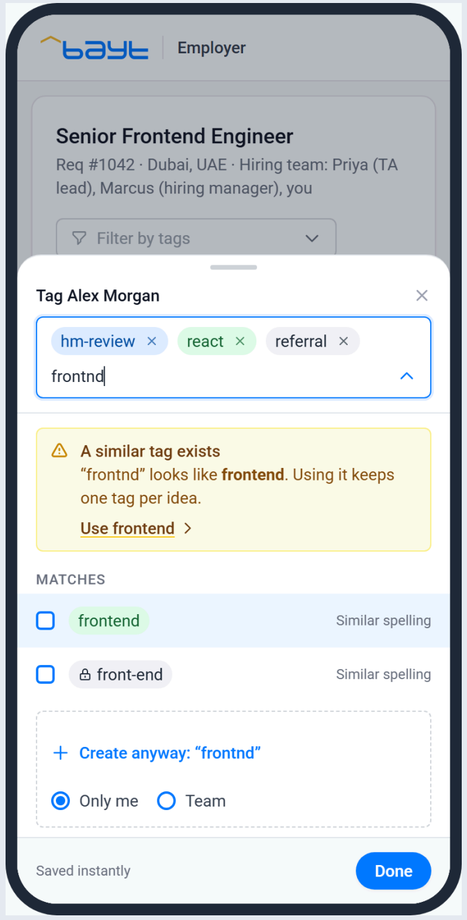
T opens the picker on the hovered row; arrows, Enter and Esc do the rest.

Scope you can see

A lock marks personal tags; everything else is shared, so nobody is surprised.

Mobile: built for capture on the move





Thumb-zone sheet

The picker rises from the bottom; targets are at least 44 px.

Suggestions over typing

Typing is slow and error-prone on phones, so "did you mean" is louder here.

Works offline

Tags apply instantly and sync when the connection returns.

Next iteration

Swipe a card to apply your most-used tag; tag from a push notification.

3 · PROTOTYPING

Consistent where it matters, adapted where it helps

Aspect	Desktop	Tablet	Mobile
Picker container	Popover	Side sheet	Bottom sheet
Main input	Keyboard: T, arrows, Enter	Touch and keyboard	Touch, 44 px targets
Bulk select	Checkboxes, shift-click	Checkboxes	Long-press to select
Tags per row	All	All, wrapped	First three, then “+N”
Navigation	Top tabs	Top tabs	Bottom bar

ALWAYS IDENTICAL

Tag colours and lock icons · suggestion order and reasons · scopes and permissions · undo · sync across devices within a second · screen-reader labels

4 · ANALYTICS & REPORTING

North Star: tag-powered recalls per weekly active user

Creating a tag is a cost. Finding something again because of a tag is the value.

Recalls per WAU
filters, chip clicks and views opened

≈

Adoption
share of WAU who tag

×

Depth
tags applied per tagger per week

×

Recall rate
recalls per tag applied

LEVERS WE CONTROL

Suggestion acceptance and time to tag raise adoption and depth. Clutter ratio and duplicate rate protect recall: a tag you can’t find again is worthless.

4 · ANALYTICS & REPORTING

Product metrics: is UTS adopted and does it work?

Metric	Definition	Target, 90 days after launch
Adoption	Share of weekly active users who apply at least one tag	Employers 35% · seekers 20%
Recall rate	Share of tagged items reopened through a tag within 30 days	40% or more
Suggestion acceptance	Share of applied tags picked from suggestions	60% or more
Time to tag	Picker opened to tag applied, median	3 seconds or less
Create ratio	New tags created per 100 tags applied	8 or fewer
Duplicate rate	Share of active tags flagged as near-duplicates	3% or less

Targets are starting points. We reset them after four weeks of beta data, segmented by persona, device and company size.

4 · ANALYTICS & REPORTING

Outcomes, performance and guardrails

Type	Metric	Target
Collaboration	Team accounts where 3+ members act on shared tags weekly	50% of team accounts
Employer outcome	Days from req opened to shortlist, taggers vs holdout	15% faster
Seeker outcome	Saved jobs that become applications, taggers vs holdout	10% higher
Performance	Suggestion response, server p95 · tag write, p95	100 ms · 200 ms
Guardrail	Tags undone or removed within one minute	10% or less
Guardrail	Search and apply conversion for all users	No drop
Guardrail	Tags hitting the sensitive-terms filter	Reviewed weekly

4 · ANALYTICS & REPORTING

One event stream, three audiences

EVENT SCHEMA

```
picker_opened  surface, targets
suggestion_shown  tags, ranks, model
tag_applied  source, rank, ms_to_tag
tag_created  scope, near_duplicate_of
tag_removed  age_ms, via_undo
tag_recall  via chip | filter | view
tag_merged · promoted · archived
every event: org, user (hashed),
persona, device, app version, time
```

Product health · weekly

For the UTS team: the metric tree, funnels, cohorts by device and persona.

Tag analytics · in product

For Priya and her VP: candidates by tag and source, time in each tag, hand-off delays.

Platform insights · monthly

For our roadmap: if hundreds of orgs invent #visa-needed, it becomes a platform filter.

4 · ANALYTICS & REPORTING

Review cadence and decision rules

Review	Who	Decides
Weekly metrics	PM, eng lead, analyst	Rollout pace, bugs, quick ranking tweaks
Fortnightly design	PM, design, research	Usability fixes from replays and feedback
Monthly taxonomy	PM, data, customer success	New platform tags, merges, group changes
Quarterly business	Leadership	OKRs, roadmap, investment level

If acceptance stays under 40% for two weeks, **then** retune ranking before widening rollout.

If the create ratio climbs past 15, **then** strengthen “did you mean” and review defaults.

If any guardrail breaks, **then** pause the rollout and roll back the flag.

5 · FEEDBACK & ITERATION

User testing in four rounds, each with an exit test

Weeks 1–3

Discover

Card sorting and tree tests with 20 users to shape groups and platform tags.

Exit: 80% sort agreement on groups

Weeks 3–5

Prototype

Moderated tests, 5–6 per persona, on all three devices: tag, recall, share, clean up.

Exit: 90% task success, ease score 6 of 7

Weeks 6–12

Beta

10 design-partner companies and 500 seekers. Two-week diary study plus in-product surveys.

Exit: product metrics trending to target

Week 13 on

Rollout

A/B at 10%, 50%, then 100%, keeping a 5% holdout for outcome metrics.

Exit: outcomes up, guardrails flat

5 · FEEDBACK & ITERATION

From feedback to a shipped change in two weeks

WHERE FEEDBACK COMES FROM

- “Not relevant” on any suggestion
- One-question survey after the fifth tag
- Session replays of the picker
- Support tickets tagged #uts
- Monthly customer advisory board
- Sales and success call notes

THE LOOP

Collect → Theme weekly → Score with RICE → Ship behind a flag → Measure, then keep or revert

Example iteration

Replays show mobile users creating #frontnd-style typos. We enlarge “did you mean” on mobile and make the existing tag the default action. Next week the mobile create ratio is compared with desktop.

6 · PERFORMANCE & SCALABILITY

Built for millions of tags without slowing down

SIZING ASSUMPTIONS

5M
tags in total

500M
taggings

100k
tags in the largest org

20 : 1
reads to writes

Planning numbers, ten times today’s expected load, to be replaced with real traffic.

HOW

- Tables and indexes partitioned by tenant, so one big org never slows others
- Prefix and fuzzy search index for type-ahead
- Each user’s top 200 tags cached in memory; most suggestions never touch the database
- Co-usage scores precomputed hourly, not per keystroke
- Bulk tagging runs as an async job with progress
- Optimistic UI: tags appear at once, sync after
- Virtualised lists and cursor paging on the client

How we prove it: targets, then five kinds of test

Service-level objective	Target
Type-ahead suggestions	p95 100 ms on the server, 150 ms on screen
Apply or remove a tag	p95 200 ms; feels instant through optimistic UI
Filter 1M items by three tags	p95 500 ms
Bulk tag 10,000 items	Done within 30 s, with progress shown

Load
Synthetic tenants with a long-tail tag mix, at 3× peak

Spike and soak
Monday-morning bulk tagging; 24-hour runs

Failure
Search index down: fall back to cache and database

Relevance
Replay real logs; precision at 5 before any ranking change

Clutter
CI check: never over 5 results, never two near-duplicates

RISKS

What could go wrong, and what we do about it

Risk	Why it matters	Mitigation
Biased or unlawful tags	Candidate tags are personal data and can encode discrimination	Block protected-characteristic terms, audit log, tags included in data access and deletion requests
Tag sprawl	Clutter kills recall and trust	Reuse-first picker, hygiene tools, create-ratio guardrail
Low adoption	Empty systems feel useless	Platform and automatic tags from day one; tagging inside existing flows
Confused with stages	Duplicated workflow and broken reports	Clear scope, stage-like tags flagged in hygiene
Migration friction	Teams already use labels and folders	Importer that maps old labels to tags and groups
Performance at scale	Slow type-ahead means nobody tags	SLOs, load tests and fallbacks from the previous slides

ROADMAP

Four phases, each gated by an exit criterion

Weeks 1–8

Personal tags MVP

Personal and platform tags on candidates and jobs. Web and mobile web. Rule-based suggestions, filters, views.

Exit: 15% beta adoption; suggest p95 under 100 ms

Weeks 9–16

Team tags

Team scope, shared views, permissions, hygiene tools, activity log, native apps.

Exit: half of beta teams use shared tags weekly

Weeks 17–24

Smart and measurable

Learned ranking, tag analytics for customers, rules, public API and webhooks.

Exit: 60% suggestion acceptance

After week 24

Platform insights

Emergent taxonomy, new platform filters, AI auto-tags that a person approves.

Exit: outcome metrics beat holdout

IN SUMMARY

One engine. A headless tag service with three scopes serves every object, device and partner.

Reuse before create. Suggestions, synonyms and hygiene keep thousands of tags usable.

Measured by recall. We count how often tags help people find things again.

Prototype: mibbrahim.github.io/unified-tagging-system

Questions?